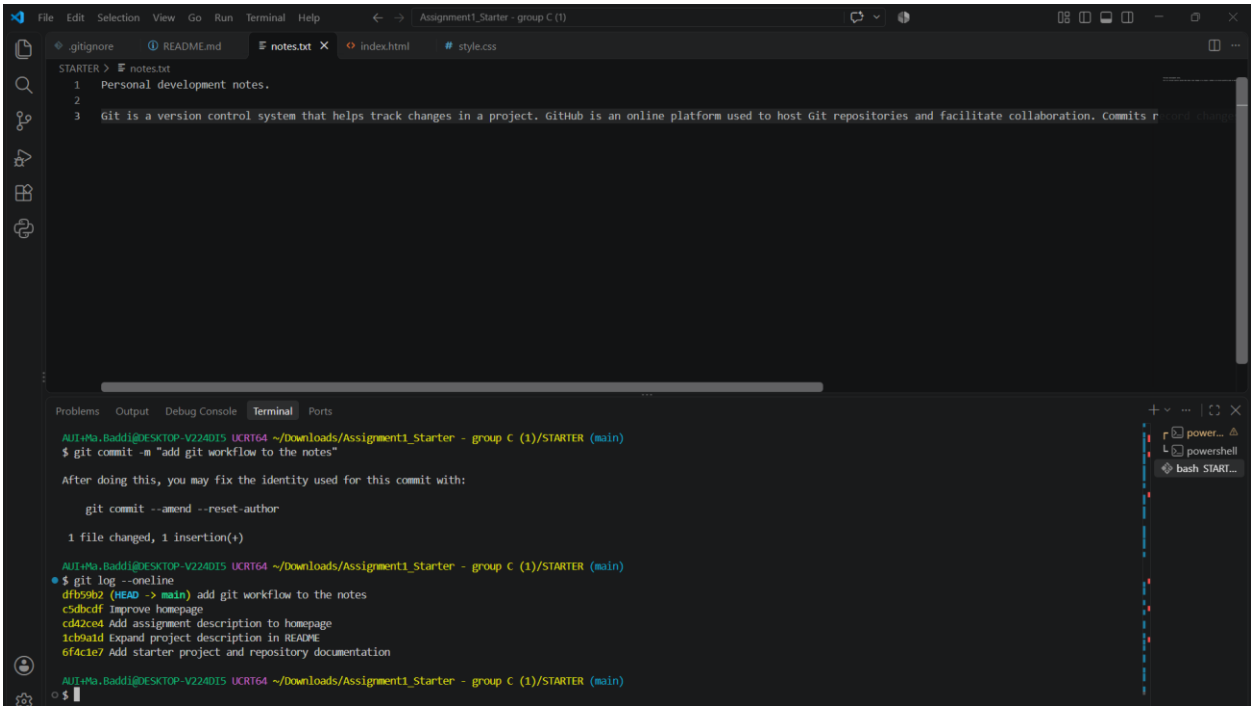
Assignment 1 - Git & GitHub Fundamentals

- **Full Name: Malak Baddi**
- **Student ID: 158934**
- **Course Code & Section: CSC 3324 03**
- **GitHub Username:m4lak-b**

1. Repository Setup

I initialized a Git repository inside the STARTER folder using git init. I created a .gitignore file to ignore local.env and regular .log files while keeping important.log visible to Git. I also created a README.md containing my name, GitHub username, project description, technologies, a code block, a link, and the provided project image.

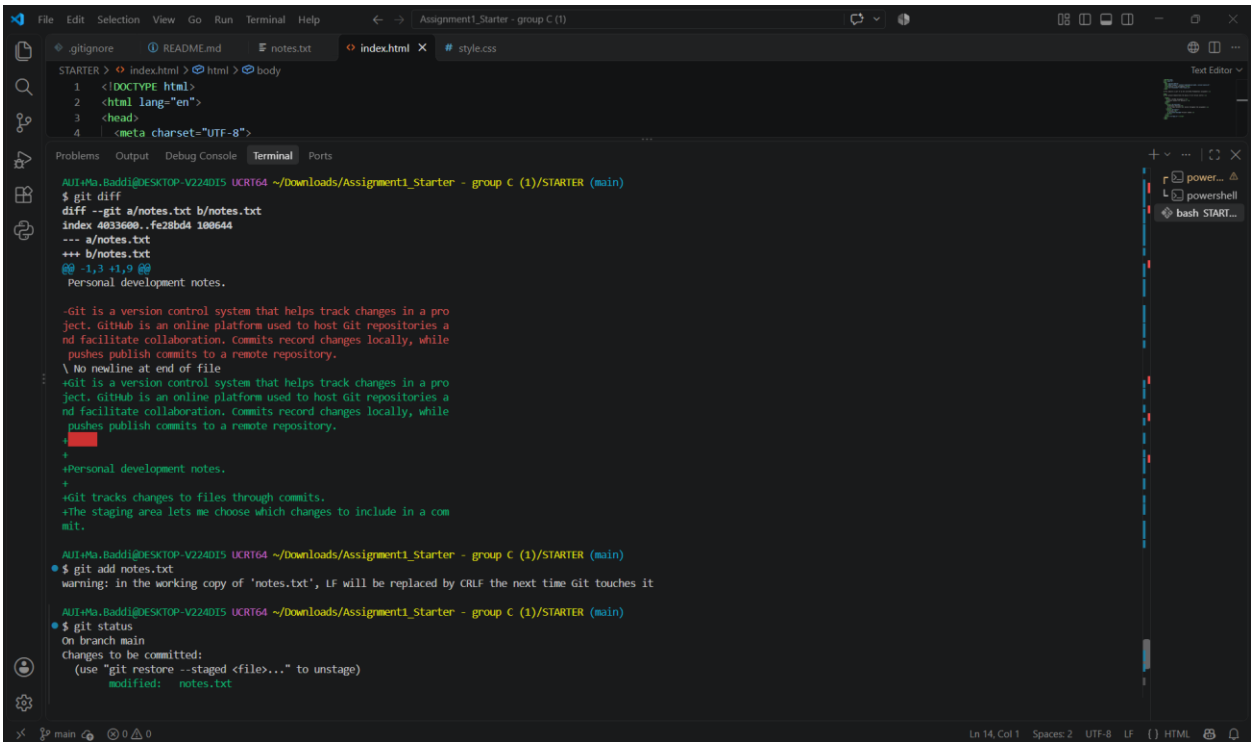
2. README & Commit History

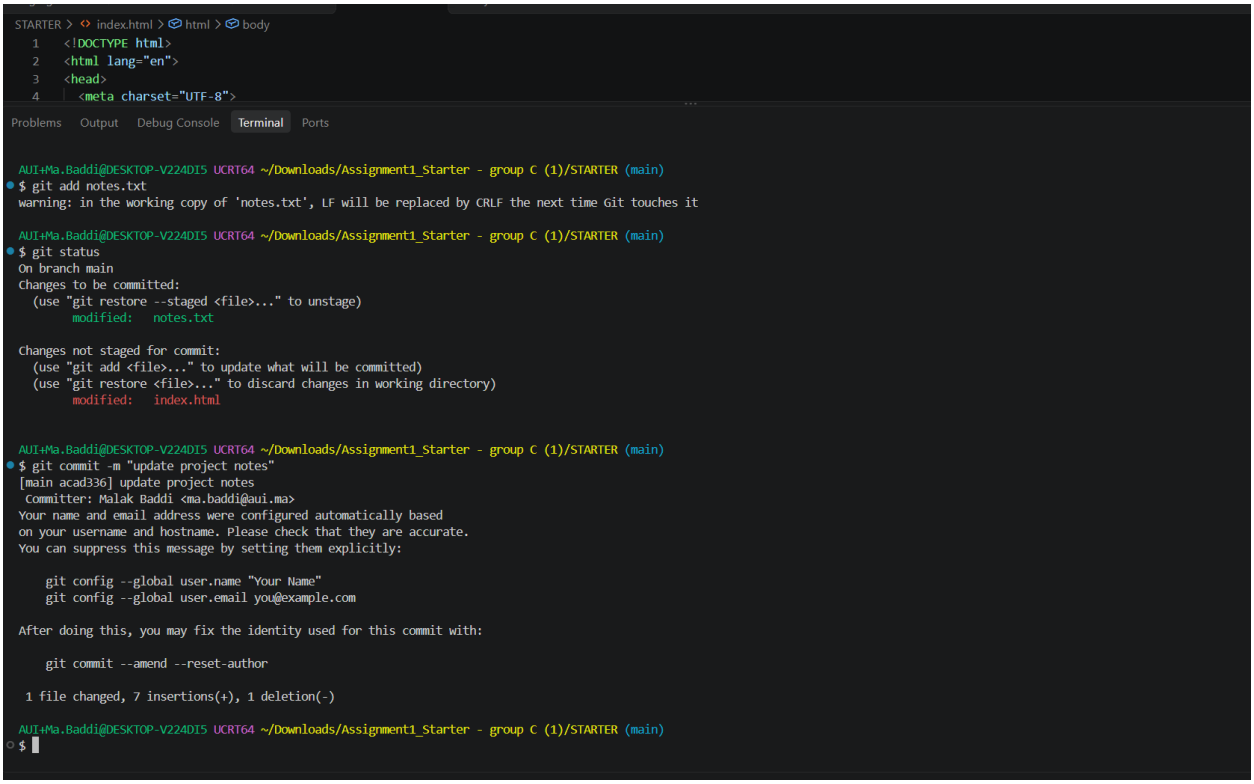


I created a README with the required project information, technologies, code example, link, and project image. I organized my work into meaningful commits covering repository setup, documentation, the homepage, styling, and project notes. I used git log --oneline to inspect the commit history.

3. Selective Staging

I modified notes.txt and index.html at the same time. I staged only notes.txt using git add notes.txt and checked git status to confirm that index.html remained unstaged. I then committed the staged file separately, demonstrating how Git allows selected changes to be included in a commit without automatically committing every modified file.

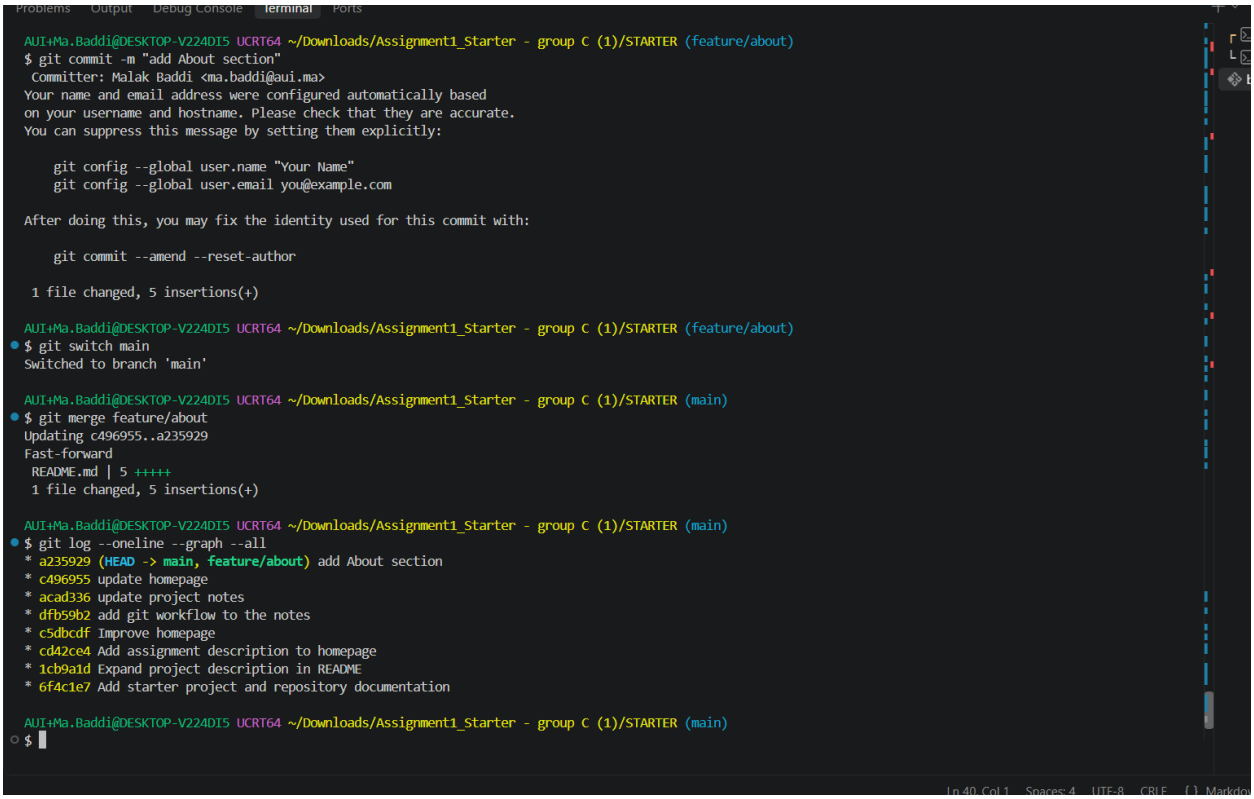




4. Branch & Merge

I created a branch named feature/about to develop an About section separately from main. I committed the About feature on that branch, switched back to main, and merged it using git merge feature/about. I used git log --oneline --graph --all to inspect the branch and

merge history.



5. Remote Changes

A commit records staged changes in the local repository, while a push uploads local commits to GitHub. Fetch downloads remote updates without integrating them into the current branch, whereas pull fetches and integrates those updates. I edited the README directly on GitHub, fetched the remote changes, and pulled them into my local main branch. I verified the synchronization using git status and inspected file changes and commit history using git diff, git diff --staged, and git log --oneline.

```
$ it status

AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/STARTER (main)
$ git status
On branch main
nothing to commit, working tree clean

AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/STARTER (main)
```

```
AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/S
$ git remote -v

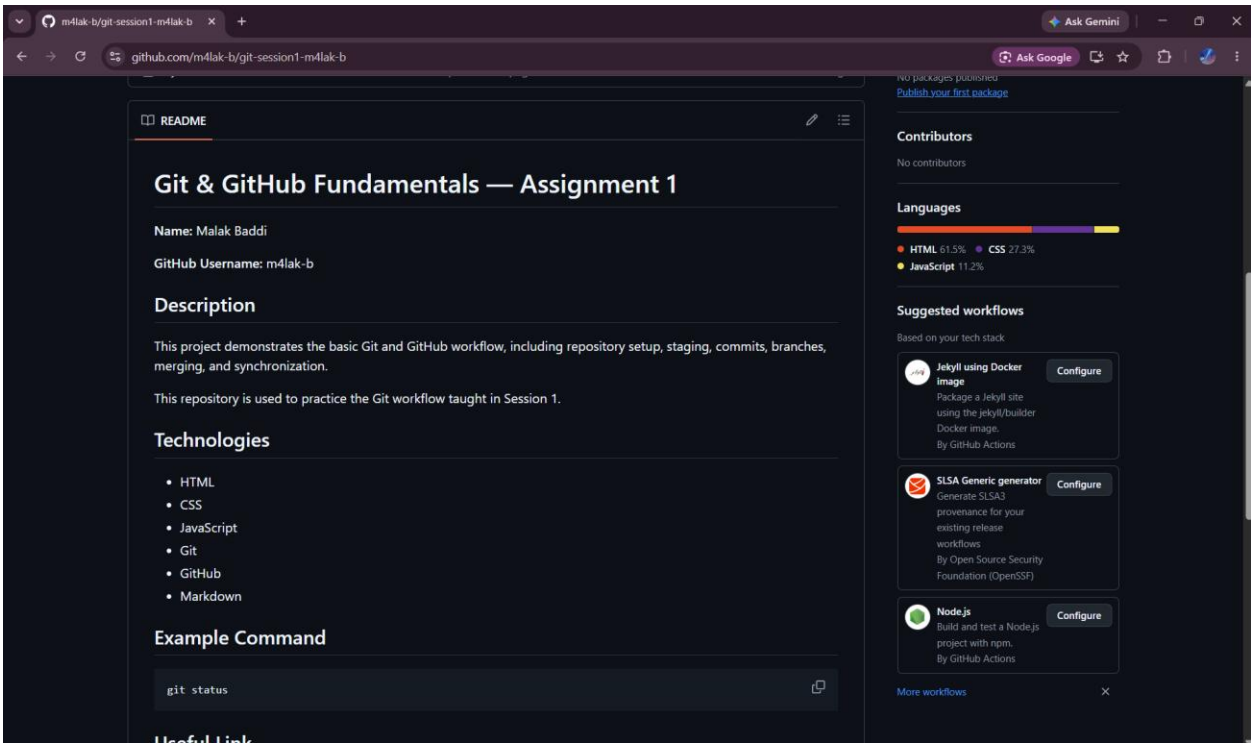
AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/S
$ git remote add origin https://github.com/m4lak-b/git-session1-m4lak-b.git

AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/S
$ git remote -v
origin  https://github.com/m4lak-b/git-session1-m4lak-b.git (fetch)
origin  https://github.com/m4lak-b/git-session1-m4lak-b.git (push)

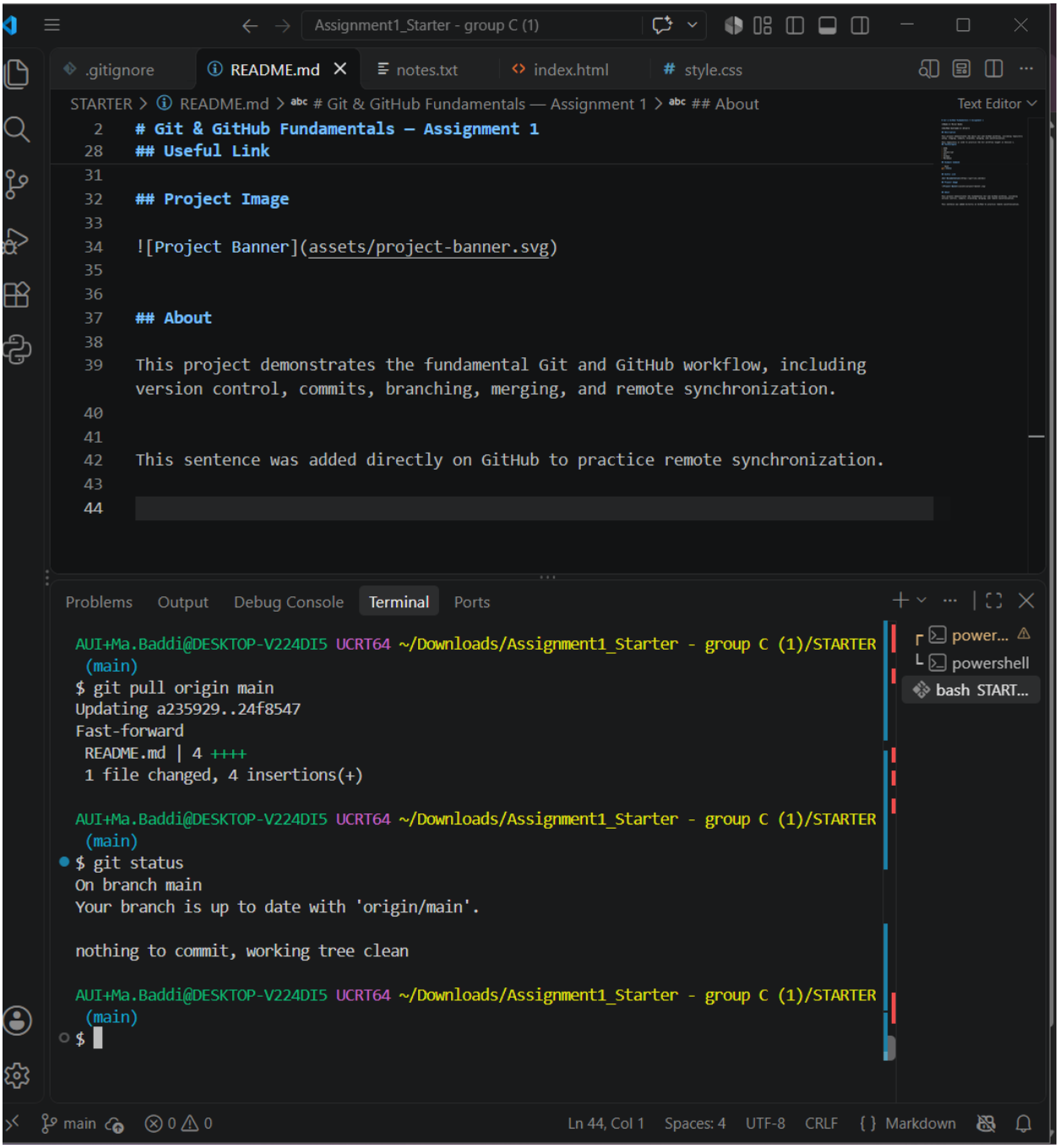
AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/S
$ git branch
  feature/about
* main

AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/S
$ git push -u origin main
Enumerating objects: 32, done.
Counting objects: 100% (32/32), done.
Delta compression using up to 22 threads
Compressing objects: 100% (29/29), done.
Writing objects: 100% (32/32), 3.82 KiB | 355.00 KiB/s, done.
Total 32 (delta 13), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (13/13), done.
To https://github.com/m4lak-b/git-session1-m4lak-b.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/S
$
```


m4lak-b/git-session1-m4lak-b
Ask Gemini
github.com/m4lak-b/git-session1-m4lak-b
Ask Google
README
Git & GitHub Fundamentals — Assignment 1
Name: Malak Baddi
GitHub Username: m4lak-b
Description
This project demonstrates the basic Git and GitHub workflow, including repository setup, staging, commits, branches, merging, and synchronization.
This repository is used to practice the Git workflow taught in Session 1.
Technologies
HTML
CSS
JavaScript
Git
GitHub
Markdown
Example Command
git status
Publish your first package
Contributors
No contributors
Languages
HTML 61.5%
CSS 27.3%
JavaScript 11.2%
Suggested workflows
Based on your tech stack
Jekyll using Docker image
Configure
Package a Jekyll site using the jekyll/builder Docker image.
By GitHub Actions
SLSA Generic generator
Configure
Generate SLSA3 provenance for your existing release workflows
By Open Source Security Foundation (OpenSSF)
Node.js
Configure
Build and test a Node.js project with npm.
By GitHub Actions
More workflows


Problems
Output
Debug Console
Terminal
Ports
AUI+Ma.Baddi@DESKTOP-V224DI5 UCRT64 ~/Downloads/Assignment1_Starter - group C (1)/STARTER (main)
$ git push -u origin main
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
$ git switch main
Already on 'main'
Your branch is up to date with 'origin/main'.
$ git fetch origin
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1007 bytes | 71.00 KiB/s, done.
From https://github.com/m4lak-b/git-session1-m4lak-b
a235929..24f8547 main -> origin/main
$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
(use "git pull" to update your local branch)
nothing to commit, working tree clean
$ git log --oneline main..origin/main
24f8547 (origin/main, origin/HEAD) Add sentence for remote synchronization practice
$
power...
powershell
bash START...



6. Fair Use of AI

I used ChatGPT as a learning tool to understand Git and GitHub commands and organize the steps required for this assignment, because I am not familiar with it. I asked questions about repository initialization, .gitignore, commits, selective staging, branches, merging, and remote synchronization. I followed the instructions in my own terminal and adapted them to the STARTER project you gave us. I verified the results using git status, git log, and the successful push and pull operations

7. Sources & Reflection

Sources

- Session 1 — Git & GitHub Fundamentals, Group C course materials.
- Git documentation: https://git-scm.com/doc
- GitHub documentation: https://docs.github.com/en
- Youtube videos

Reflection

The most challenging concept was understanding the difference between the working directory, staging area, local repository, and remote repository. If Git behaves unexpectedly, I would first use git status to inspect the current branch and identify staged, unstaged, or untracked changes. I would then use git diff and git log to understand the changes and history before deciding what to do next.